

Mumbai University MSc IT Sem II Modern Networking Master Guide

Exam-slip-first solved practical pack

This guide is written for a beginner who has never opened Packet Tracer before and needs a stepwise, exam-safe path from **question slip** → **topology** → **wiring** → **configuration** → **verification**.

How to use this file

For every practical, follow the same rhythm:

1. Read the **question**.
2. Build the **topology in Packet Tracer** or, for advanced topics, in **GNS3/EVE-NG**.
3. Make the **connections** exactly as written.
4. Configure the devices in the order shown.
5. Verify with the commands listed at the end.
6. Save the file and take screenshots for your journal.

One important reality check

Some topics in the Mumbai University Modern Networking practicals are easy in Packet Tracer, while some are not truly supported there.

Usually fine in Packet Tracer

- Static routing
- IP SLA basics
- PBR basics
- BGP basics
- Inter-VLAN routing
- Secure management plane
- Simple AS_PATH demonstrations

Usually better in GNS3 or EVE-NG

- MPLS

- VRF / MPLS VPN
- SDN
- OpenFlow
- Some advanced BGP policy labs

So this guide gives you:

- **Packet Tracer instructions** wherever PT is good enough
 - **GNS3/EVE-NG instructions** where Packet Tracer starts to wobble like a shopping cart with one broken wheel
-

Quick Packet Tracer beginner map

Where to click

- **Bottom-left:** Network Devices
- **Routers icon:** choose routers like 1941, 2811, or 2911
- **Switches icon:** choose 2960 or 3560
- **End Devices:** PCs, Laptops, Servers
- **Connections icon:** lightning bolt

Common cables

- **PC to Switch:** Copper Straight-Through
- **Router to Switch:** Copper Straight-Through
- **Router to Router over Ethernet:** Copper Cross-Over or Straight-Through if auto-MDIX is available
- **Router to Router over Serial:** Serial DCE / DTE

If you need serial ports

Many router models do not have serial interfaces by default.

1. Click the router
2. Open the **Physical** tab
3. Turn power **off**
4. Add a serial module such as **WIC-2T** or **HWIC-2T**
5. Turn power **on**

If one side of the serial link is DCE, set:

clock rate 64000

Basic IOS skeleton

```
enable
configure terminal
hostname R1
interface g0/0
ip address 10.0.0.1 255.255.255.0
no shutdown
end
write memory
```

1) Configure IP SLA Tracking and Path Control Topology

This is the classic branch-to-ISP failover practical from your manual. The manual shows R1 as the branch router, with two ISP paths and loopbacks for testing. The configuration below follows the same idea. The topology in the manual is on page 3 and the configuration starts on page 4. The running results and route changes are shown in pages 5 to 13.

What the question means

You are building a branch router that prefers one ISP, but can move to another route when the tracked path fails.

Devices to drag

- 3 routers
- 1 PC or server if your examiner wants an end host
- Optional: one cloud or server icon to represent the Internet side

Suggested Packet Tracer devices

- Routers: **2911**
- End device: **PC**
- If serial ports are missing, add serial modules first

How to place them

Put:

- **R1** at the bottom as the branch router
- **R2** on the left as ISP1
- **R3** on the right as ISP2

How to wire them

Use **Serial DCE** cables.

Connections:

- R1 `s1/0` to R2 `s1/0`
- R1 `s1/1` to R3 `s1/0`
- R2 `s1/1` to R3 `s1/1` if your diagram uses the Internet link between ISP1 and ISP2

If your topology uses different serial numbers, keep the same logic:

- one link from branch to ISP1
- one link from branch to ISP2
- optionally one link between ISP1 and ISP2

IP plan used in the manual style

- R1 loopback0: `192.168.1.1/24`
- R1 to R2: `209.165.201.0/30`
- R1 to R3: `209.165.202.128/30`
- R2 loopback0: `209.165.201.30`
- R3 loopback0: `209.165.202.158`

Step 1: Configure interface IP addresses

R1

```
enable
conf t
hostname R1

interface loopback0
 ip address 192.168.1.1 255.255.255.0
 exit

interface s1/0
 ip address 209.165.201.2 255.255.255.252
 no shutdown
 exit

interface s1/1
 ip address 209.165.202.130 255.255.255.252
 no shutdown
 exit
```

R2

```
enable
conf t
hostname R2

interface loopback0
 ip address 209.165.201.30 255.255.255.255
 exit

interface s1/0
 ip address 209.165.201.1 255.255.255.252
 no shutdown
 exit
```

R3

```
enable
conf t
hostname R3

interface loopback0
```

```
ip address 209.165.202.158 255.255.255.255
exit
```

```
interface s1/0
ip address 209.165.202.129 255.255.255.252
no shutdown
exit
```

Step 2: Add a default route on R1

First make the default route with a normal static route:

```
ip route 0.0.0.0 0.0.0.0 209.165.201.1
```

Step 3: Create IP SLA object

On R1, probe ISP1 DNS or loopback address:

```
ip sla 12
icmp-echo 209.165.201.30
frequency 11
exit
```

```
ip sla schedule 12 life forever start-time now
```

Step 4: Add tracking

```
track 1 ip sla 12 reachability
delay down 10 up 1
exit
```

Step 5: Bind the tracked route

Now replace the plain default route with a tracked one:

```
no ip route 0.0.0.0 0.0.0.0 209.165.201.1
ip route 0.0.0.0 0.0.0.0 209.165.201.1 5 track 1
```

If you want a backup route, use a higher administrative distance on the alternate path.

Step 6: Verify

```
show ip sla configuration 12
show ip sla statistics
show track 1
show ip route
traceroute 209.165.200.254 source 192.168.1.1
```

What the examiner wants to see

- IP SLA is configured
- Track object is up
- Default route changes when the tracked path breaks
- `show ip route` shows the active route
- `show ip sla statistics` shows success and timeout changes

Common mistakes

- Forgetting `no shutdown`
- Using the wrong serial interface number
- Forgetting to set `clock rate` on the DCE side
- Probing the wrong target IP

Public .pkt note

For this topic, you can search GitHub for Packet Tracer labs that include IP SLA and static failover. Exact Mumbai University .pkt files were not found in the public search I ran, so use the closest matching lab and rebuild the addressing.

2) Using the AS_PATH Attribute

This practical is about using BGP path selection and AS_PATH filtering. The exam slip and manual both point to this topic. Your slip file shows the question name and the journal manual has the classroom version of the same theme.

What the question means

You must:

- build a BGP topology
- use AS_PATH to influence or filter routes
- avoid leaking private AS numbers where needed

Beginner topology in writing

Use 3 routers:

- **R2** = Andheri, AS 100
- **R1** = Bandra, AS 300
- **R3** = Churchgate, AS 65000

Connections:

- R2 s1/0 ↔ R1 s1/0
- R1 s1/1 ↔ R3 s1/1

Loopbacks:

- R2 loopback0 = 10.1.1.1/24
- R1 loopback0 = 10.2.2.2/24
- R3 loopback0 = 10.3.3.1/24

Step 1: Interface configuration

R2

```
enable
conf t
hostname R2
interface s1/0
```



```
ip address 192.168.1.2 255.255.255.0
no shutdown
exit
interface loopback0
ip address 10.1.1.1 255.255.255.0
exit
```

R1

```
enable
conf t
hostname R1
interface s1/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
interface s1/1
ip address 172.24.1.1 255.255.255.0
no shutdown
exit
interface loopback0
ip address 10.2.2.2 255.255.255.0
exit
```

R3

```
enable
conf t
hostname R3
interface s1/1
ip address 172.24.1.3 255.255.255.0
no shutdown
exit
interface loopback0
ip address 10.3.3.1 255.255.255.0
exit
```

Step 2: Configure BGP

R2, AS 100

```
router bgp 100
 network 10.1.1.0 mask 255.255.255.0
 neighbor 192.168.1.1 remote-as 300
```

R1, AS 300

```
router bgp 300
 network 10.2.2.0 mask 255.255.255.0
 neighbor 192.168.1.2 remote-as 100
 neighbor 172.24.1.3 remote-as 65000
```

R3, AS 65000

```
router bgp 65000
 network 10.3.3.0 mask 255.255.255.0
 neighbor 172.24.1.1 remote-as 300
```

Step 3: AS_PATH filtering

Simple route filtering with AS-path access-list

Example:

- accept only routes that came from AS 100

```
ip as-path access-list 10 permit ^100$
route-map ONLY_FROM_AS100 permit 10
 match as-path 10
```

Apply it on the receiving neighbor:

```
router bgp 300
 neighbor 192.168.1.2 route-map ONLY_FROM_AS100 in
```

Prevent private AS numbers from being advertised

On the edge router that talks outside, use:

```
router bgp 300
neighbor 172.24.1.3 remove-private-as
```

If your IOS image or Packet Tracer version does not accept `remove-private-as`, explain it in viva and show the AS-path filter idea with a route-map.

Step 4: Verification

```
show ip bgp
show ip bgp summary
show ip route
show ip bgp regexp ^100$
ping 10.3.3.1 source 10.1.1.1
```

What to say in viva

- AS_PATH is the sequence of AS numbers a route has crossed
- Shorter AS_PATH is usually preferred when other BGP attributes are equal
- AS_PATH can be matched with `ip as-path access-list`
- Route maps can use that match to permit or deny routes

Public .pkt note

Search GitHub for:

- `bgp.pkt`
- `cisco packet tracer bgp`
- `AS_PATH packet tracer`

A directly visible `.pkt` file exists in at least one public repository: `prakashkm03/Packet_Tracer` contains `bgp.pkt`. A broader Packet Tracer BGP example also exists on GitHub and can be used as a reference lab.

3) Configuring IBGP and EBGP Sessions, Local Preference, and MED

This is the big one. The exam slip explicitly asks for IBGP and EBGP, and another version asks for Local Preference and MED too.

What the question means

You need to:

- build inside-AS IBGP
- build outside-AS EBGP
- use `next-hop-self`
- prefer one path using **Local Preference**
- influence external route choice using **MED**

Beginner topology in writing

Use 3 routers:

- **R1** = internal router 1, AS 64512
- **R2** = internal router 2, AS 64512
- **ISP** = external router, AS 200

Connections:

- R1 to ISP over one serial link
- R2 to ISP over one serial link
- R1 to R2 either with an Ethernet link or loopback-based iBGP

If you want the easiest exam build, use loopback iBGP between R1 and R2 and serial EBGP to ISP.

Suggested addressing

- R1 loopback0: `172.16.64.1/24`
- R2 loopback0: `172.16.32.1/24`
- ISP loopback0: `10.2.2.1/24`
- R1 to ISP: `192.168.1.4/30`

- R2 to ISP: 192.168.1.0/30

Step 1: Configure interfaces

R1

```
enable
conf t
hostname R1
interface loopback0
  ip address 172.16.64.1 255.255.255.0
exit
interface s1/1
  ip address 192.168.1.5 255.255.255.252
  no shutdown
exit
```

R2

```
enable
conf t
hostname R2
interface loopback0
  ip address 172.16.32.1 255.255.255.0
exit
interface s1/1
  ip address 192.168.1.1 255.255.255.252
  no shutdown
exit
```

ISP

```
enable
conf t
hostname ISP
interface loopback0
  ip address 10.2.2.1 255.255.255.0
exit
interface s1/0
```

```
ip address 192.168.1.6 255.255.255.252
no shutdown
exit
interface s1/1
ip address 192.168.1.2 255.255.255.252
no shutdown
exit
```

Step 2: Configure eBGP to ISP

R1

```
router bgp 64512
network 172.16.64.0 mask 255.255.255.0
neighbor 192.168.1.6 remote-as 200
```

R2

```
router bgp 64512
network 172.16.32.0 mask 255.255.255.0
neighbor 192.168.1.2 remote-as 200
```

ISP

```
router bgp 200
network 10.2.2.0 mask 255.255.255.0
neighbor 192.168.1.5 remote-as 64512
neighbor 192.168.1.1 remote-as 64512
```

Step 3: Configure iBGP between R1 and R2

Use loopbacks so the session stays alive even if a link is replaced later.

R1

```
router bgp 64512
  neighbor 172.16.32.1 remote-as 64512
  neighbor 172.16.32.1 update-source loopback0
  neighbor 172.16.32.1 next-hop-self
```

R2

```
router bgp 64512
  neighbor 172.16.64.1 remote-as 64512
  neighbor 172.16.64.1 update-source loopback0
  neighbor 172.16.64.1 next-hop-self
```

Step 4: Local Preference

Local Preference tells your AS which exit path to prefer. Higher is better.

Example: prefer R1 as the primary exit.

On R1

```
route-map PRIMARY_T1_IN permit 10
  set local-preference 160
exit

router bgp 64512
  neighbor 192.168.1.6 route-map PRIMARY_T1_IN in
```

On R2

```
route-map SECONDARY_T1_IN permit 10
  set local-preference 125
exit

router bgp 64512
  neighbor 192.168.1.2 route-map SECONDARY_T1_IN in
```

Step 5: MED

MED is advertised to the neighbor and suggests which incoming path should be preferred.

Lower MED is better.

On R1

```
route-map PRIMARY_T1_MED_OUT permit 10
  set metric 50
exit

router bgp 64512
  neighbor 192.168.1.6 route-map PRIMARY_T1_MED_OUT out
```

On R2

```
route-map SECONDARY_T1_MED_OUT permit 10
  set metric 75
exit

router bgp 64512
  neighbor 192.168.1.2 route-map SECONDARY_T1_MED_OUT out
```

Step 6: Verification

```
show ip bgp summary
show ip bgp
show ip route
clear ip bgp * soft
```

Check that:

- the higher local preference path wins inside the AS
- the lower MED path is preferred from the neighbor side
- `next-hop-self` prevents unreachable next hops in iBGP

Viva one-liners

- **iBGP** is BGP between routers in the same AS
- **eBGP** is BGP between different ASes
- **Local Preference** controls outbound path choice in the local AS
- **MED** influences how a neighboring AS enters your AS
- **next-hop-self** changes the next hop to the local router

Public .pkt note

Search GitHub for:

- `Packet Tracer BGP`
- `bgp.pkt`
- `IBGP EBGp Packet Tracer`

A public `.pkt` BGP example exists in `prakashkm03/Packet_Tracer/blob/master/bgp.pkt` .
For deeper policy behavior, Packet Tracer may be too limited, so use the config section here as your exam reference.

4) Secure the Management Plane

The exam slip asks for this directly. Your journal also has this practical with secure access, username/password security, AAA RADIUS, and remote management.

What the question means

Lock down the router so only trusted users can reach it.

Devices to drag

- 1 router
- 1 switch
- 1 PC
- Optional: 1 Server if you want RADIUS

How to wire

- PC to switch: Copper Straight-Through
- Switch to router: Copper Straight-Through
- If using server: server to switch with Straight-Through

Step 1: Basic secure access

```
enable
conf t
hostname R3
security passwords min-length 10
enable secret class12345

line console 0
password ciscoconpass
exec-timeout 5 0
login
logging synchronous
exit

line vty 0 4
password ciscovtypass
exec-timeout 5 0
login
exit

line aux 0
no exec
exit

service password-encryption
banner motd $Unauthorized access strictly prohibited!$
```

Step 2: Username and local login

```
username JR-ADMIN secret class12345
username ADMIN secret class54321
```

```
line console 0
  login local
  exit

line vty 0 4
  login local
  exit
```

Step 3: SSH remote management

If your examiner wants secure remote access, this is the good gold:

```
ip domain-name mu.local
crypto key generate rsa
ip ssh version 2

line vty 0 4
  transport input ssh
  login local
  exit
```

Step 4: AAA RADIUS with local fallback

If your lab includes a server acting as RADIUS:

```
aaa new-model

radius server RADIUS-1
  address ipv4 192.168.1.101
  key RADIUS-1-pa55w0rd
  exit

radius server RADIUS-2
  address ipv4 192.168.1.102
  key RADIUS-2-pa55w0rd
  exit

aaa group server radius RADIUS-GROUP
  server name RADIUS-1
  server name RADIUS-2
```

```
exit
```

```
aaa authentication login default group RADIUS-GROUP local  
aaa authentication login TELNET-LOGIN group RADIUS-GROUP local
```

```
line vty 0 4  
login authentication TELNET-LOGIN  
transport input ssh telnet  
exit
```

If Packet Tracer does not let you fully emulate RADIUS, explain the AAA concept in viva and show the local fallback plus SSH.

Verification

```
show running-config  
show ip ssh  
show users  
show aaa servers
```

What to say in viva

- `enable secret` protects privileged exec mode
- `service password-encryption` hides line passwords in config output
- `login local` uses local usernames
- SSH is preferred over Telnet because it encrypts the session
- AAA centralizes authentication

Public .pkt note

Search for security labs with SSH, AAA, and management-plane hardening. Packet Tracer security labs are common, but exact Mumbai University `.pkt` files were not publicly verified in the search I ran.

5) Configure and Verify Path Control Using PBR

PBR is a favorite exam practical. Your manual includes a full version with route-map policy routing and verification.

What the question means

Normally routers pick paths using routing protocols. PBR lets you say:
“for traffic from this source, send it that way.”

Beginner topology in writing

Use 4 routers:

- R1 at the left as source LAN
- R2 top-left path
- R3 top-right path
- R4 right-side destination LAN

Connections:

- R1 ↔ R2
- R1 ↔ R3
- R2 ↔ R3
- R3 ↔ R4

Suggested device placement

- R1 bottom-left
- R2 top-left
- R3 center-right
- R4 far-right

Interfaces and cables

Use serial links or Ethernet links depending on your router model. Serial is simpler to explain in viva.

If using serial:

- choose **Serial DCE** links
- set clock rate on DCE side

Addressing idea

Use four networks:

- R1 to R2: 172.16.12.0/24
- R1 to R3: 172.16.13.0/24
- R2 to R3: 172.16.23.0/24
- R3 to R4: 172.16.34.0/29

Loopbacks:

- R1 192.168.1.1/24
- R2 192.168.2.1/24
- R3 192.168.3.1/24
- R4 192.168.4.1/25 and 192.168.4.129/25

Step 1: Configure interfaces

R1

```
enable
conf t
hostname R1
interface s1/0
  ip address 172.16.12.1 255.255.255.0
  bandwidth 128
  no shutdown
exit
interface s1/1
  ip address 172.16.13.1 255.255.255.0
  bandwidth 64
  no shutdown
exit
interface loopback0
  ip address 192.168.1.1 255.255.255.0
exit
```

R2

```
enable
conf t
hostname R2
interface s1/0
  ip address 172.16.12.2 255.255.255.0
  bandwidth 128
  no shutdown
exit
interface s1/1
  ip address 172.16.23.2 255.255.255.0
  bandwidth 128
  no shutdown
exit
interface loopback0
  ip address 192.168.2.1 255.255.255.0
exit
```

R3

```
enable
conf t
hostname R3
interface s1/0
  ip address 172.16.13.3 255.255.255.0
  bandwidth 64
  no shutdown
exit
interface s1/1
  ip address 172.16.23.3 255.255.255.0
  bandwidth 128
  no shutdown
exit
interface s1/2
  ip address 172.16.34.3 255.255.255.248
  no shutdown
exit
interface loopback0
  ip address 192.168.3.1 255.255.255.0
exit
```

R4

```
enable
conf t
hostname R4
interface s1/0
  ip address 172.16.34.4 255.255.255.248
  no shutdown
exit
interface loopback4
  ip address 192.168.4.1 255.255.255.128
exit
interface loopback5
  ip address 192.168.4.129 255.255.255.128
exit
```

Step 2: Add basic routing

You may use static routing or a dynamic protocol. The practical journal uses PBR on top of route reachability, so at minimum ensure all routers know the route to the destination LAN.

For quick exam safety, add static routes or OSPF.

Example:

```
ip route 192.168.4.0 255.255.255.128 172.16.34.4
ip route 192.168.4.128 255.255.255.128 172.16.34.4
```

Step 3: Create ACL for the traffic to steer

Example: send traffic from 192.168.5.0/24 via a chosen next hop.

```
access-list 1 permit 192.168.5.0 0.0.0.255
```

Step 4: Create route-map

```
route-map r3-to-r1 permit 10
  match ip address 1
  set ip next-hop 172.16.13.1
exit
```


Step 5: Apply PBR on the receiving interface

On the interface that receives the traffic from the source network:

```
interface s1/2
  ip policy route-map r3-to-r1
exit
```

Step 6: Verification

```
show route-map
show ip policy
show access-lists
traceroute 192.168.1.1 source 192.168.4.1
traceroute 192.168.1.1 source 192.168.5.1
```

What the examiner wants

- Policy route-map created
- ACL matches the right source
- `set ip next-hop` is working
- traceroute shows the chosen route changes based on policy

Public .pkt note

Look for Packet Tracer PBR labs or rebuild this topology manually. The manual and journal already give you the core commands, so a public `.pkt` is optional rather than essential here.

6) Inter-VLAN Routing

The slip and the journal both include Inter-VLAN Routing. This one is very Packet Tracer friendly.

What the question means

Different VLANs should talk to each other through:

- a router-on-a-stick setup, or
- a Layer 3 switch with SVIs

For a beginner, router-on-a-stick is the easiest to understand.

Beginner topology in writing

Use:

- 1 router
- 1 switch
- 2 PCs

PC1 in VLAN 10

PC2 in VLAN 20

How to wire

- PC1 to switch `fa0/1` with Straight-Through
- PC2 to switch `fa0/2` with Straight-Through
- Router `g0/0` to switch `fa0/24` with Straight-Through

Step 1: Create VLANs on the switch

```
enable
conf t
vlan 10
  name SALES
exit
vlan 20
  name HR
exit
```

Step 2: Assign access ports

```
interface fa0/1
  switchport mode access
  switchport access vlan 10
exit
```

```
interface fa0/2
  switchport mode access
  switchport access vlan 20
exit
```

Step 3: Make trunk to router

```
interface fa0/24
  switchport mode trunk
exit
```

Step 4: Router-on-a-stick

On the router:

```
enable
conf t
hostname R1
interface g0/0
  no shutdown
exit

interface g0/0.10
  encapsulation dot1Q 10
  ip address 10.1.110.1 255.255.255.0
exit

interface g0/0.20
  encapsulation dot1Q 20
  ip address 10.1.120.1 255.255.255.0
exit
```

Step 5: Set PC IPs

- PC1: 10.1.110.10/24 , gateway 10.1.110.1
- PC2: 10.1.120.10/24 , gateway 10.1.120.1

Step 6: Verification

```
show vlan brief
show interfaces trunk
ping 10.1.120.10
```

If you use a Layer 3 switch instead

Use `ip routing` , create SVI interfaces, and assign VLANs to ports. The journal contains examples with Layer 3 switching and EtherChannel that point in this direction.

7) Simulating MPLS Environment

This is where Packet Tracer starts to get a little grumpy. For MPLS, use **GNS3** or **EVE-NG**.

What the question means

You must build a service-provider core with MPLS label switching, usually using:

- PE routers
- P routers
- CE routers
- OSPF in the core
- MPLS LDP on the core

Beginner topology in writing

A standard 5-router layout:

- R1 and R3 are PE routers
- R2 is a P router in the core
- R4 and R5 are CE routers

Connections:

- R1 ↔ R2
- R2 ↔ R3
- R1 ↔ R4
- R3 ↔ R5

Step 1: Core addressing

Assign IP addresses to the core links using /30 networks.

Step 2: OSPF in the core

Run OSPF area 0 between R1, R2, R3.

```
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
```

Step 3: Enable MPLS on core links

```
mpls ip
mpls label protocol ldp
```

On interface level:

```
interface g0/0
 ip address 10.0.0.1 255.255.255.252
 mpls ip
 no shutdown
```

Step 4: CE side

CE routers get their own LAN and static/OSPF connectivity to the PE routers.

Step 5: Verification

```
show mpls ldp neighbor
show mpls forwarding-table
show ip route
traceroute 1.1.1.1 source 10.0.1.3
```

Honest note

Packet Tracer does not fully emulate MPLS the way real lab environments do. Use GNS3 or EVE-NG here, because the emulator is the actual beast that can carry this backpack.

Public .pkt note

For MPLS, public Packet Tracer files are usually not real MPLS labs. Search for GNS3 or EVE-NG labs instead.

8) Simulating VRF

Your manual and the slip both point to VRF. In reality, this is also better in GNS3 or EVE-NG. The manual's VRF lab is an MPLS VPN-style design with PE and CE routers.

What the question means

Create separate routing tables for different customers or sites so their traffic stays logically isolated.

Beginner topology in writing

Use:

- R1 and R3 as PE routers
- R2 as core/P router
- R4 and R5 as CE routers on two sites

The manual's task list says:

- build IGP in the core under OSPF area 0
- run MPLS LDP in the core
- create VRF-A1 on site 1 and VRF-A2 on site 2

- use RD and RT 500:1
- use OSPF between PE and CE
- use VPNv4 between PE routers
- redistribute OSPF and BGP under VRF

Step 1: Create VRF on PE routers

On R1

```
ip vrf A-1
  rd 500:1
  route-target both 500:1
exit
```

On R3

```
ip vrf A-2
  rd 500:1
  route-target both 500:1
exit
```

Step 2: Assign CE-facing interfaces to VRF

R1

```
interface e0/1
  ip vrf forwarding A-1
  ip address 192.168.1.1 255.255.255.0
  no shutdown
exit
```

R3

```
interface e0/1
  ip vrf forwarding A-2
  ip address 192.168.2.1 255.255.255.0
```

```
no shutdown
exit
```

Step 3: Run OSPF between PE and CE

Example on R1:

```
router ospf 10 vrf A-1
 network 192.168.1.0 0.0.0.255 area 0
exit
```

Example on CE router R4:

```
router ospf 10
 network 192.168.1.0 0.0.0.255 area 0
exit
```

Step 4: Configure MP-BGP VPNv4 between PE routers

```
router bgp 500
 no bgp default ipv4-unicast
 neighbor 3.3.3.3 remote-as 500
 neighbor 3.3.3.3 update-source loopback0
 address-family vpnv4 unicast
  neighbor 3.3.3.3 activate
  neighbor 3.3.3.3 send-community extended
  neighbor 3.3.3.3 next-hop-self
exit-address-family
```

Step 5: Redistribute between OSPF and BGP under VRF

On R1

```
router bgp 500
 address-family ipv4 vrf A-1
  redistribute ospf 10 vrf A-1 match internal external 1 external 2
exit-address-family
```



```
exit

router ospf 10 vrf A-1
 redistribute bgp 500 subnets
exit
```

On R3

Use the same pattern for VRF A-2.

Step 6: Verification

```
show ip route vrf A-1
show ip route vrf A-1 bgp
show ip bgp vpnv4 all
ping vrf A-1 4.4.4.4
traceroute 2.2.2.2 source 10.0.1.3
```

Public .pkt note

This is not a Packet Tracer-friendly feature set. Use GNS3/EVE-NG. The manual itself notes the first two tasks in the VRF lab are preconfigured in the MPLS LDP lab.

9) Design the following topology and simulate static routing

The slip file includes a static routing practical. This is the easiest one on the list.

What the question means

You are manually pointing routers toward the next hop. No dynamic routing protocol is involved.

Beginner topology in writing

Use 3 routers:

- R1 in the middle or branch side
- R2 on one side
- R3 on the other side

Or use the exact topology in your slip. The principle stays the same.

How to wire

Use serial or Ethernet links between the routers.

For beginners, serial is easier to understand in exam diagrams.

Step 1: Configure interface IPs

Example:

R1

```
interface s1/0
  ip address 10.1.1.2 255.255.255.252
  no shutdown
exit
interface s1/1
  ip address 10.2.2.1 255.255.255.252
  no shutdown
exit
```

R2

```
interface s1/0
  ip address 10.1.1.1 255.255.255.252
  no shutdown
exit
```

R3

```
interface s1/0
  ip address 10.2.2.2 255.255.255.252
  no shutdown
exit
```

Step 2: Add static routes

```
ip route 192.168.2.0 255.255.255.0 10.1.1.1
ip route 192.168.3.0 255.255.255.0 10.2.2.2
```

Step 3: Verify

```
show ip route
ping 192.168.2.1
ping 192.168.3.1
traceroute 192.168.3.1
```

What to remember

- Static routes are hand-written directions
- Next hop must be reachable
- Wrong subnet mask means wrong road map

Public Packet Tracer / lab sources appendix

I searched public repositories and Packet Tracer examples to find ready-made labs. I did **not** find a publicly verified exact Mumbai University .pkt for every slip, but these are useful starting points:

- [prakashkm03/Packet_Tracer/blob/master/bgp.pkt](#)
- [rmj3197/Cisco-Packet-Tracer-Project/blob/master/Router%20Config%20using%20OSPF.pkt](#)
- [SameetAsadullah/Static-Routing-Using-Cisco-Packet-Tracer](#)
- [Rakibul73/Cisco_Network_Routing_Switching](#)
- [r/Cisco Packet Tracer](#) style topic repositories and labs on GitHub

Search terms that work well:

- `bgp pkt cisco packet tracer`
 - `ip sla packet tracer`
 - `pbr packet tracer`
 - `inter vlan packet tracer`
 - `static routing pkt`
 - `secure management plane cisco packet tracer`
-

Viva speed-run answers

IP SLA

Tracks reachability by periodically probing a target.

Tracking

Links an object like a route to a live condition.

AS_PATH

List of AS numbers a BGP route has crossed.

iBGP

BGP inside the same AS.

eBGP

BGP between different ASes.

Local Preference

Controls the preferred outbound path in the local AS. Higher wins.

MED

Suggests which entry path another AS should prefer. Lower wins.

PBR

Forces packets to follow policy instead of normal routing logic.

VRF

Creates separate routing tables on the same router.

MPLS

Forwarding with labels instead of only IP lookup.

Inter-VLAN routing

Lets VLANs talk through a router or Layer 3 switch.

Final practical checklist

Before the exam:

- learn how to add serial modules
- learn how to open CLI on router and switch
- know the cable types
- memorize `show ip route`, `show ip bgp`, `show ip sla statistics`, `show route-map`, `show interfaces trunk`
- save every file with `write memory`

The trick is not wizardry. It is repetition with clean wiring and clean configs. Once the topology clicks, the rest is just command choreography.